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Cosolvents induce sharp softening and RNA-like twist-stretch coupling of RNA-DNA hybrid

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ABSTRACT Cellular environments generally contain abundant organic cosolvents, yet how in-cell cosolvents modulate the mechanics of the RNA-DNA hybrid (RDH) duplex remains unclear. Here, we combined high-precision magnetic tweezer experiments with all-atom molecular dynamics simulations to investigate the structural elasticity of RDH duplexes using polyethylene glycol (PEG)200, PEG1000, and ethanol as model cosolvents. We found that high-concentration cosolvents sharply soften RDH duplexes in bending and induce an apparent RNA-like positive twist-stretch coupling. Specifically, for the RDH duplex, the bending persistence length P decreases from ∼54 to ∼27–35 nm (∼35%–50% decrease) by 40% cosolvents. Meanwhile, the twist-stretch coupling parameter dL/dN shifts from ∼0 to −0.56 to −0.45 nm/turn, approaching that of the RNA duplex (∼−0.81 nm/turn). Microscopic analyses revealed that sharp bend softening mainly arises from local hydrogen-bond disruption associated with structural mismatch between RNA and DNA strands and enhanced strand fluctuation driven by increased ion neutralization. Moreover, the RNA-like twist-stretch coupling originates from cosolvent-induced shifts in the structure and deformation pathway of the RDH duplex toward RNA-like ones, as evidenced by comprehensive analyses at the helix, base-pair, and strand levels. Our findings revealed that cosolvents can substantially reshape the structure and mechanics of RDH duplexes, potentially providing a physical basis for understanding their behavior in cells.

SIGNIFICANCE The structure and mechanics of RNA-DNA hybrid (RDH) duplexes play important roles in their biological functions, and RDHs generally reside in cellular environments enriched with organic cosolvents, while how in-cell cosolvents regulate the structural elasticity of RDHs remains unclear. In this work, we combined high-precision magnetic tweezer (MT) experiments with MD simulations to explore the elasticity of RDH duplexes, with polyethylene glycol (PEG)200, PEG1000, and ethanol as model cosolvents. Our results revealed that the cosolvents sharply soften RDH duplexes in bending by ∼35%–50% and induce a pronounced RNA-like positive twist-stretch coupling. This study demonstrated that cosolvents can substantially reshape the structure and elasticity of RDH duplexes, providing insight into RDH behavior in the presence of cosolvents and guidance for RDH-based nanostructure design.

INTRODUCTION

RNA-DNA hybrid (RDH) duplexes are fundamental intermediates that arise in diverse cellular processes, including transcription bubbles, Okazaki fragmentation during DNA replication, R-loop formation, and CRISPR-Cas-mediated nucleic acid recognition.[1–5] Existing experiments suggested that the structure and mechanics of RDH duplexes can be crucial for their biological functions and clinical applications.[2,3,6] For example, the ASO (antisense oligonucleotide) technique is beneficial for gene expression-related diseases through RNase H-mediated degradation of RNA within RDH duplexes.[6] Structurally, RDH duplexes adopt an intermediate helical conformation between the canonical A-form RNA and B-form DNA helices, and each strand in RDH duplexes maintains its essential dynamics in pure DNA or RNA duplexes, i.e., the DNA strand is more flexible and the RNA strand is more rigid in RDH duplexes.[7–9] Single-molecule techniques such as magnetic tweezers (MT) and all-atom molecular dynamics (MD) simulations have been

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employed to characterize the mechanics of RDH duplexes in different dimensions, as well as the effects of ions and temperature.[7–12]

Under near-physiological ion conditions, an RDH duplex exhibits intermediate mechanical properties between A-form RNA and B-form DNA duplexes. Specifically, an RDH duplex displays an intermediate persistence length (P_{RDH}) of ∼50–58 nm and an intermediate stretching modulus (S) of ∼600–800 pN, which are between the corresponding values of RNA and DNA duplexes (P_{RNA} ∼54–60 nm and P_{DNA} ∼45–50 nm; S_{RNA} ∼350–600 pN and S_{DNA} ∼1,000–1,500 pN).[7–10,13–27] Furthermore, the RDH duplex displays a negligible twist-stretch coupling (g ∼0), which is an intermediate value between those of RNA and DNA duplexes since the g of RNA and DNA duplexes are strikingly opposite.[7,9,16,28,29] Moreover, P and S of RDH duplexes show a similar monovalent salt dependence with those of RNA and DNA duplexes due to the structural similarity of the duplexes.[9,30,31] Additionally, the elastic parameters of the RDH duplex share a similar temperature dependence with those of RNA and DNA duplexes.[10] However, like RNAs and DNAs, RDHs generally reside in cellular environments consisting of high concentrations of biomolecules as cosolvents, while the structural mechanics of RDH duplexes remain unclear in cellular environments.[32–41]

Polyethylene glycols (PEGs) of various molecular weights and ethanol have been widely used as model cosolvents in exploring the structure and stability of nucleic acids in cellular environments.[32,35–37,42,43] For example, PEG200 at high concentration (e.g., 40%) has usually been employed to derive nearest-neighbor thermodynamic parameters for DNA, RNA, and RDH base pairing/stacking in cellular environments.[32,35,37] Furthermore, high concentrations of PEG200 have also been reported to destabilize DNA, RNA, and RDH duplexes, whereas larger PEGs such as PEG8000 can slightly stabilize them.[32,33,36,44] Moreover, PEG1000 has been used as a cosolvent to explore RNA tertiary structure folding in cellular environments.[45,46] Additionally, ethanol of high concentrations can induce a B-to-A transition for DNA duplexes[47,48] and has been applied to explore the structure of human telomeric DNA in cosolvent environments.[49] However, despite these advances, the effect of cosolvents such as PEGs and ethanol on the structures and mechanics of RDH duplexes still remains poorly understood.

In this work, we combined high-precision MT experiments with all-atom MD simulations to explore how cosolvents modulate the structure and mechanics of the RDH duplex. The cosolvents used include PEG200, PEG1000, and ethanol, covering a wide range of volume concentrations of [0%–40%]. We found that high concentrations of PEG200, PEG1000, and ethanol all cause a sharp bend softening and an RNA-like positive twist-stretch coupling of the RDH duplex, revealing the distinct structure and mechanics of RDH duplexes in the cosolvent environments.

MATERIALS AND METHODS

High-precision MT measurements for the mechanics of RDH duplex

To explore the structure and mechanics of RDH duplexes in cosolvent environments, we conducted high-precision MT measurements for RDH duplexes in diverse model cosolvents including PEG200, PEG1000, and ethanol, covering concentrations of [0%–40%]. Firstly, for tensile elasticity assays, the 13,751-bp RDH duplex was prepared following the protocol described by Zhang et al.[9] Briefly, the two complementary single-stranded (ss) RNA and ssDNA strands, where the ssRNA was transcribed from the 13,751-bp DNA duplex containing a T7 promoter using T7 RNA polymerase and the ssDNA was generated through one-sided PCR (OSP) using an SH-labeled short DNA fragment as primer, were annealed together to generate an RDH duplex with 3′ overhangs at one end. A biotin-labeled short DNA fragment was annealed to the 3′ overhangs of the RDH duplex to generate the final product, which contains a 13,704-bp RDH and a 47-bp DNA duplex. In the twist-stretch coupling measurements, the torsion-constrained 13,751-bp RDH duplex was constructed according to the method of Zhang et al.[9] More details about the construction of the RDH duplex can be found in the supplemental information (Figure S1). Different concentrations of PEG200 and ethanol solutions were diluted in 5 mM Tris-HCl (pH 7.5). Furthermore, different concentrations of PEG1000 were dissolved in 5 mM Tris-HCl. NaCl was dissolved in PEG or ethanol solution to reach a final concentration of 150 mM. The NaCl solution without PEGs or ethanol was used as a control.

For tensile elasticity assays, one end of an RDH molecule was anchored to a superparamagnetic microbead, with the opposite end anchored to a coverslip (Figure 1A). A pair of magnets was used to apply forces to the anchored RDH molecule.[50] More details about the MT measurements can be found in our previous studies.[16,51–53] To quantify the changes in the mechanical properties of the RDH duplex, the force extension curves were fitted to the worm-like chain (WLC) model using the following formula:[54,55]

$$F = \frac{k_B T}{P}\left[0.25\left(1 - \frac{L}{L_c}\right)^{-2} + \frac{L}{L_c} - 0.25\right]. \quad (1)$$

The measured variables, L and F, represent the molecular extension and the applied tensile force, respectively. The two fitted elastic parameters are the contour length (L_c) and the bending persistence length (P) (see Figure 2A).

Notably, the fitting curves are in agreement with the extensions obtained from experiments at forces below 1 pN under low ethanol and PEG concentrations. Nevertheless, to maintain methodological consistency and eliminate low-force artifacts, we selected the 1–20 pN force range for WLC fitting across all cosolvent conditions. An exception is at

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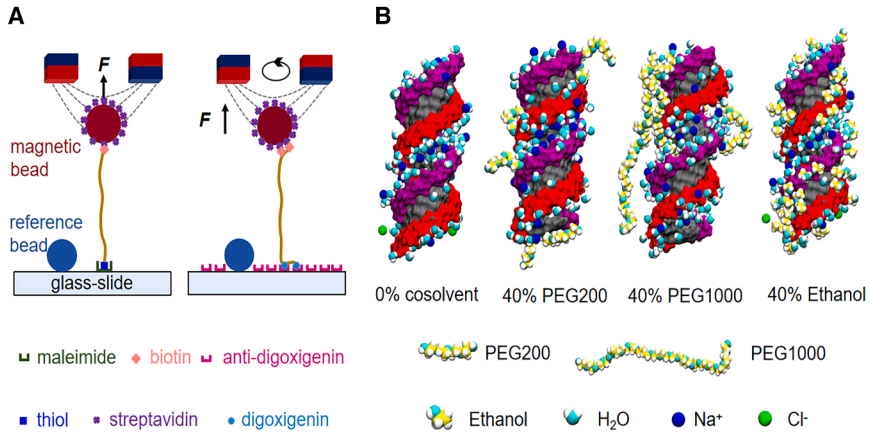


Figure 1. Our MT experiments and all-atom MD simulations for measuring the structures and elasticities of the RNA-DNA hybrid (RDH) duplex. (A) The home-built magnetic tweezer. A pair of NdFeB magnets is used to stretch the RDH duplex anchored between a glass slide and a microbead (left). A rotational motor system was utilized to manipulate the rotation and position of the magnets along the *z* axis (right). (B) Schematic representation of the initial structure of 20-bp RDH duplex without cosolvents and with cosolvents including PEG200, PEG1000, and ethanol in our MD simulations.

the high concentrations (e.g., 30% and 40%) of PEG200 and PEG1000, where the RDH undergoes condensation (Figures S2 and S3). To avoid RDH condensation in our measurements, we fitted these data to the higher-force range of 2–20 pN, as higher stretching forces would counteract interactions for the condensation and maintain extended RDH conformations. Such RDH condensation may come from the effects of dehydration and volume exclusion by high

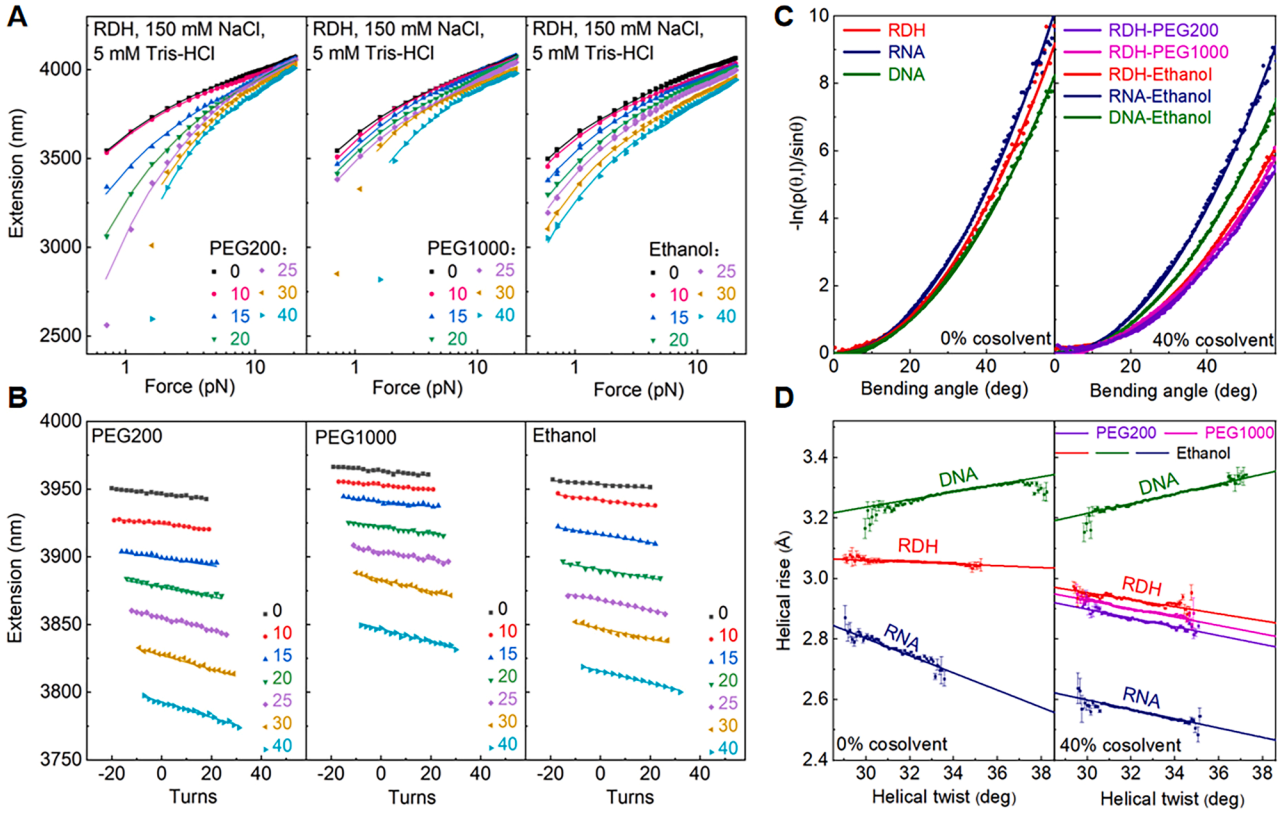


Figure 2. Effects of cosolvent on the elastic properties of the RDH duplex from our MT measurements and MD simulations. (A) Representative force-extension (*F*-*x*) curves of RDH duplex at various PEG200, PEG1000, and ethanol concentrations. The *F*-*x* curve is fitted to Equation 1 (solid line), yielding a bending persistence length *P*. (B) Representative extensions (*L*) versus rotation turns (*N*) were obtained around the torsional relax point of the RDH duplex at 8 pN using MT. *L* was fitted to linear functions of *N* (solid lines). (C) The bending angle distributions $p(\theta)$ versus bending angle θ for the RDH duplex from MD trajectories without cosolvents and with 40% cosolvents, including PEG200, PEG1000, and ethanol. (D) The relationships between helical rise and helical twist for RDH duplex without cosolvent and with 40% cosolvent, including PEG200, PEG1000, and ethanol. The error bars denote the standard errors.

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cosolvents, as the former could amplify the cation-RDH interactions through reducing the effective dielectric constant and the latter could provide a spatial confinement for conformation transition through volume exclusion.[56]

For twist-stretch coupling assays, the multi-biotinylated end of the RDH molecule was anchored onto a streptavidin-coated superparamagnetic microbead, while its multi-digoxigenin-labeled end was anchored to an anti-digoxigenin-functionalized coverslip. The rotation-extension curve was measured at 8 pN in the range of about ± 20 turns surrounding the torsion relaxed point using a rotational motor. The torsion relaxed point was determined at 0.3 pN. The twist-stretch coupling value was calculated via linear fitting of the rotation-extension curves; see Figure 2B for the rotation-extension curves at typical cosolvents. Please see more details of twist-stretch coupling measurements in our previous studies[16,29,53,57] and in the supplemental information.

Molecular simulation calculations for the mechanics of RDH duplex

All-atom MD simulations

To understand the structure and mechanics of the RDH duplex in cosolvent environments and the underlying mechanisms, we performed a series of all-atom MD simulations of an RDH duplex in the presence of cosolvents including PEG200, PEG1000, and ethanol. Firstly, we built a 20-bp RDH duplex with the DNA sequence of 5′-CGACTCTACGG CATCTGCGC-3′ and its complementary RNA using the University of California, San Francisco, Chimera program.[58] The sequence was selected based on previous experiments and MD simulations for short nucleic acid helices.[29,59] Our MD simulations were carried out using the Gromacs 2021.5 software package[60] with AMBER bsc1+χOL3 force fields for the RDH duplex,[61–63] an explicit TIP3P water model[64] for molecular interactions, and the optimized potentials for liquid simulations (OPLS) force field for PEGs and ethanol.[65,66] The force fields have been shown to well describe the static and dynamic properties of the RDH duplex,[61–63] water molecules,[64] and PEGs/ethanol,[65,66] respectively. The systems were immersed in a rectangular periodic box containing the RDH duplex, water molecules, cosolvent molecules, Na^+ ions to neutralize the negatively charged RDH duplex, and additional Na^+ and Cl^- ions to mimic the physiological concentration of 150 mM NaCl.[67] The simulation boxes containing the water-cosolvent mixtures,[65] as shown in Figure 1B, were built following the well-established protocol described by Cheatham et al.[47] Specifically, the initial systems were constructed with the RDH duplex surrounded by water, in which ions and cosolvent molecules were placed at least 20 Å away from the surface of the RDH duplex.

In order to reveal how cosolvents affect the elasticity of the RDH duplex, we (1) used different typical cosolvents including PEG200, PEG1000, and ethanol and (2) covered the broad concentration range from 0% to 40% in volume percentage (vol/vol) for PEG200, PEG1000, and ethanol, as it was reported that cells are generally occupied by macromolecules as cosolvents of 10%–40% in volume.[33,44,68] The numbers of cosolvent and water molecules were calculated from their volume percentages of the initial simulation box based on the corresponding densities and molar masses, which are listed in Table S1.

After the systems were built, we first performed energy minimization to eliminate any possible clashes and bad contacts. Then, an isothermal-isochoric (NVT) ensemble was conducted for 2 ns to heat the systems to 298 K through V-rescale temperature coupling. Afterward, an isothermal-isobaric (NPT) ensemble was conducted to maintain pressure at 1 atm over 8 ns with Parrinello-Rahman pressure coupling, ensuring the solvents reached equilibrium. During the equilibration period, we carried out a harmonic potential with a force constant of 1,000 kJ/mol $\cdot$ nm^{-2} to restrain the RDH duplex. A cutoff of 1 nm was used for short-range Coulomb interactions and the particle mesh Ewald method, as well as periodic boundary conditions for long-range electrostatic interactions.[69] All bonds containing hydrogen bonds were constrained using the linear constraint solve (LINCS) algorithm.[70] After the pre-equilibration, we employed an unrestrained production run of 800 ns for each system and 2 fs as an integration step in conjunction with the leapfrog. According to Figure S4, the RDH duplex reached equilibrium at ∼300 ns, and the last 500 ns MD trajectories were used for the analyses. Our MD simulations also showed that the number of cosolvents within typical radial distances from the RDH duplex reached a stable plateau, indicating an equilibrium solvent distribution around the duplex, as shown in Figure S5. Moreover, we verified the volume percentage of added cosolvents by calculating the equilibrium volume percentage of added cosolvents based on the equilibrated box dimensions (see Table S1). Additionally, we performed 800-ns MD simulations for the RNA and DNA duplexes with ethanol added at different concentrations as the references. Please see our recent works[14,16,28,29] and the supplemental information for more details.

Calculations of helical parameters and elastic parameters for RDH duplex

Based on the MD trajectories, we analyzed the major helical parameters and central helical axis of the RDH duplex using Curves+,[71] including helical rise, helical twist, widths of the major and minor grooves, depths of the major and minor grooves, inclination, and helical radius.[7,13,16,28,29] In our analyses, we removed three base pairs at each end of the RDH duplex to avoid the end effect.[7,13,72] Besides, we also calculated the rigid body parameters and the torsion angles, including sugar pucker angles, backbone torsion, and intra-strand phosphorus-phosphorus (P-P) distance to describe the

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geometry of the RNA and DNA strands in the RDH duplex.[7,16]

Based on the WLC model, the bending persistence length P of the RDH duplex can be calculated through fitting the probability $p(\theta)$ to the following equation:[14,24,73]

$$-\ln\left[\frac{p(\theta)}{\sin\theta}\right] = \frac{P\theta^2}{2L}, \quad (2)$$

where θ is the bending angle over the used segment of contour length L and $p(\theta)$ is the probability distribution of θ; see Figure 2C for the $p(\theta)$ of the RDH duplex at typical cosolvents. Here, θ was calculated as the angle between the tangents of the two end base stacks along the central axis determined with Curves+.[71] Moreover, we also calculated the twist-stretch coupling g, which is associated with length L and cumulative helical twist Φ. In analogy to the previous works[16,29] and our MT measurements, we use dL/dN to describe the twist-stretch coupling of the RDH duplex in the MD calculations, where dN and dL are the change in helical turn and the change in extension of the RDH duplex, respectively; see Figure 2D for the $L \sim N$ curves of the RDH duplex at typical cosolvents. It is noted that dL/d$N > 0$ and dL/d$N < 0$ correspond to the negative and positive coupling, respectively, since dL/d$N \propto -g/S$, where g and S are the twist-stretch coupling and the stretch modulus with positive values.[7,13,29]

Calculations of ion binding, hydrogen bonds, and solvent dielectric constant for RDH duplex

To understand the structural mechanics of the RDH duplex in the presence of cosolvents, we analyzed ion binding and hydrogen bonding within the RDH duplex in the presence of the various cosolvents, including PEG200, PEG1000, and ethanol. In analogy to our previous work,[14–16,29] we firstly calculated the binding ions around the RDH duplex, which was defined as those ions within a helical distance of 11 Å from the central axis and the ions within a radial distance of 6 Å from phosphate atoms (see Figure S6). Moreover, we calculated the distance of hydrogen bonds between base pairs of the RDH duplex.[59] In addition, since the addition of cosolvents such as PEGs and ethanol would alter ion-RNA interactions, we calculated the dielectric constant of the solutions with the used cosolvents using the built-in command of "gmx current" in the Gromacs software[60] based on the MD trajectories.

RESULTS AND DISCUSSION

MT measurements on the mechanical properties of RDH duplex in cosolvent environments

To explore the structural mechanics of the RDH duplex in cosolvent environments, we measured the bending persistence lengths P, contour lengths L_c, and twist-stretch coupling parameters dL/dN for the RDH duplex in the presence of the cosolvents, including PEG200, PEG1000, and ethanol, over a broad concentration range of [0%–40%] through high-precision MT experiments (see Figure 1A and materials and methods). For each cosolvent concentration, we repeated the MT measurements for at least five molecules to obtain average values and corresponding standard deviations of P, L_c, and dL/dN for the RDH duplex.

Firstly, we measured the force-extension (F-x) curves for the RDH duplex in solutions containing various cosolvents, including PEG200, PEG1000, and ethanol of different concentrations, through high-precision MT experiments (see materials and methods). Through fitting each F-x curve to Equation 1 based on the extensible WLC model,[54,74] we obtained the bending persistence length P and contour length L_c for the RDH duplex in solutions with different cosolvents. As shown in Figure 3A, the addition of different cosolvents leads to a sharp bend softening for the RDH duplex, as reflected by a substantial decrease (>∼35%) in its bending persistence length. Specifically, P significantly decreases from ∼54.3 to ∼27.0 nm by ∼50% for 40% PEG200, to ∼31.0 nm by ∼43% for 40% PEG1000, and to ∼35.4 nm by ∼35% for 40% ethanol, respectively, which are significantly greater than the slight decreases in P for the RNA (∼18%) and DNA (∼13%) duplexes caused by 40% ethanol from our previous MT experiments, as shown in Figure 3B.[16] Moreover, the contour length L_c of the RDH duplex slightly decreases with the addition of different cosolvents, e.g., 40% cosolvents shorten the RDH duplex by ∼6% for 40% PEG200, by ∼2% for 40% PEG1000, and by ∼5% for 40% ethanol, as shown in Figures 3C and 3D.

Secondly, we quantified the twist-stretch coupling of the RDH duplex using high-precision MT experiments, which describes the correlation between deformations in twist and stretch dimensions. The twist-stretch coupling parameters dL/dN of the RDH duplex in the presence of different cosolvents were derived through the extension-turns curves from the MT measurements (see materials and methods). As shown in Figure 3E, the addition of 40% cosolvents visibly decreases dL/dN of the RDH duplex from ∼−0.16 to ∼−0.56 nm/turn for PEG200, to ∼−0.48 nm/turn for PEG1000, and to ∼−0.45 nm/turn for ethanol, and such values are visibly of the same sign as that (∼−0.81 nm/turn) of the RNA duplex.[16,28,29,53] Furthermore, Figure 3F shows that the 40% cosolvents decrease dL/dN of the RDH duplex, in contrast to our previous MT measurements that 40% ethanol increases dL/dN for both RNA and DNA duplexes.[16] It is rather interesting that the RDH duplex exhibits an RNA-like positive twist-stretch coupling (dL/d$N < 0$) in the cosolvent environments with different cosolvents, while the twist-stretch coupling of the RDH duplex is negligible (dL/dN ∼0) in the absence of cosolvents. It is noted that dL/d$N > 0$ and dL/d$N < 0$ correspond to negative and positive coupling, respectively[7,13,29] (see materials and methods).

Therefore, our high-precision MT experiments reveal the surprising findings that an RDH duplex exhibits sharply

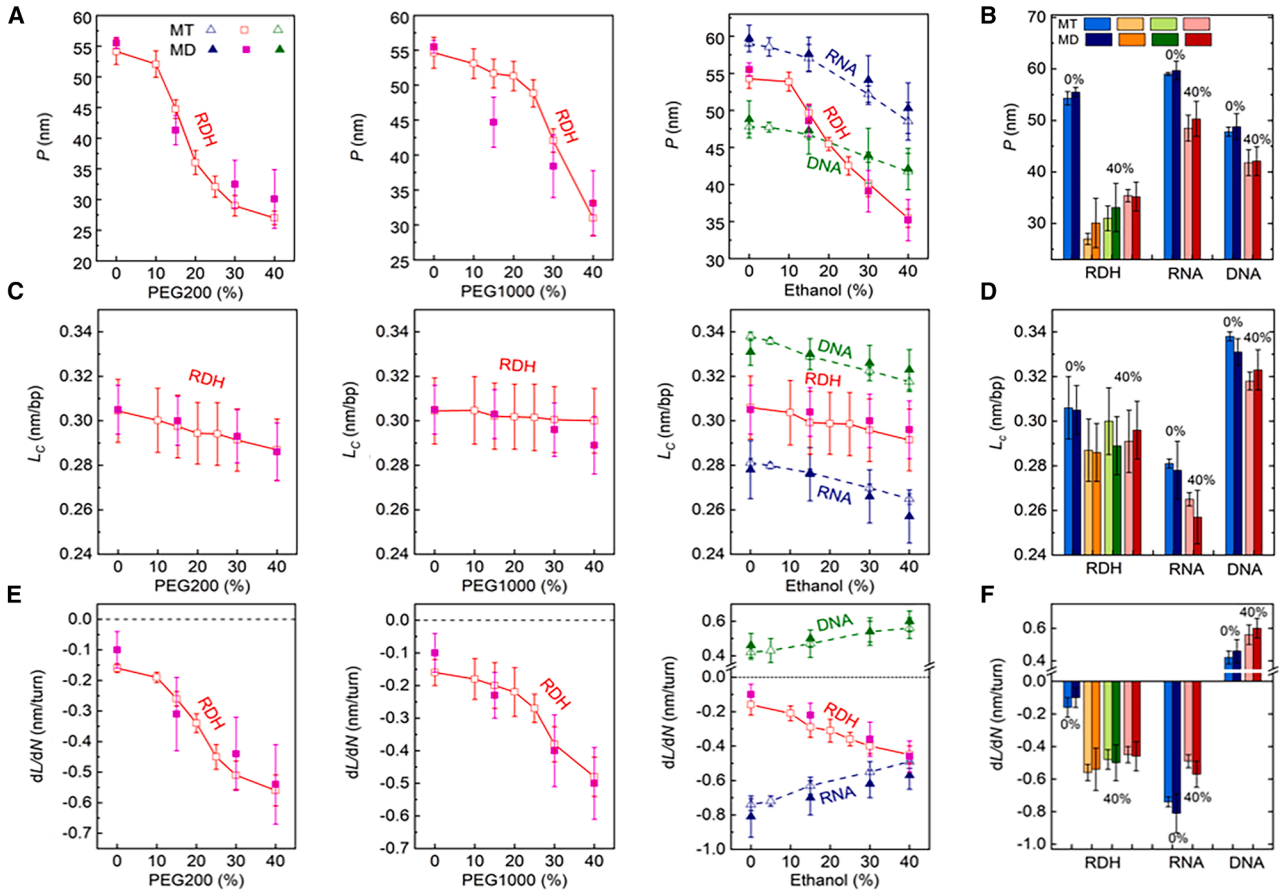


Figure 3. Bending persistence length P, contour length L_c, and twist-stretch coupling parameter dL/dN of RDH in the presence of different cosolvents, including PEG200, PEG1000, and ethanol, from MT measurements and MD simulations. (A) P as a function of cosolvent concentration for PEG200, PEG1000, and ethanol. The error bars denote the standard deviations around the mean values from MT measurements using at least five molecules or the mean values of five equal intervals in our MD trajectories. (B) Comparison of P for the RDH duplex at 0% and 40% cosolvents from both MT measurements and MD simulations. Light and dark bars represent the MT and MD results, and orange, green, and red bars correspond to PEG200, PEG1000, and ethanol, respectively. (C) Contour length as a function of cosolvent concentration for PEG200, PEG1000, and ethanol. (D) Comparison of L_c with the same descriptions as (B). (E) dL/dN as a function of cosolvent concentration for PEG200, PEG1000, and ethanol. (F) Comparison of dL/dN with the same descriptions as (B). Here, the values of RNA and DNA duplexes with added ethanol from MT results are shown as references.[16]

softened bending and an apparently RNA-like positive twist-stretch coupling in cosolvent environments, compared with the markedly higher bending stiffness and negligible twist-stretch coupling of the RDH duplex in the absence of cosolvents.[7,9,16,28,29,53]

MD calculations on the mechanical properties of RDH duplex in cosolvent environments

To understand the underlying mechanisms of the surprising effect of cosolvents on the mechanical properties of the RDH duplex, we performed extensive all-atom MD simulations of the RDH duplex in the presence of the different cosolvents, including PEG200, PEG1000, and ethanol, over a broad concentration range from 0% to 40%. In analogy to the MT measurements, we calculated the bending persistence length P, contour length L_c, and twist-stretch coupling parameter dL/dN for the RDH duplex.

Firstly, we calculated the bending persistence length P of the RDH duplex based on the probability distributions of bending angle θ for the RDH duplex from the MD trajectories (see materials and methods). In accordance with the MT measurements, our MD simulations revealed that a high concentration of cosolvents leads to a sharp bend softening for the RDH duplex. Specifically, the cosolvents of 40% concentration cause sharp decreases in P from ~55.5 to ~30.1 nm for PEG200 by ~46%, to ~33.1 nm for PEG1000 by ~40%, and to ~35.4 nm for ethanol by ~36%, respectively, which are significantly greater than the decreases in P for the RNA duplex from 59.7 to 50.3 nm (~16%) and for the DNA duplex from 48.8 to 42.1 nm (~14%) caused by 40% ethanol from the MD simulations, as shown in Figures 3A and 3B. Moreover, our MD calculations also showed that PEG200, PEG1000, and ethanol slightly shorten the RDH duplex by ~6%, ~5%, and ~3% in contour length, L_c, as the cosolvent

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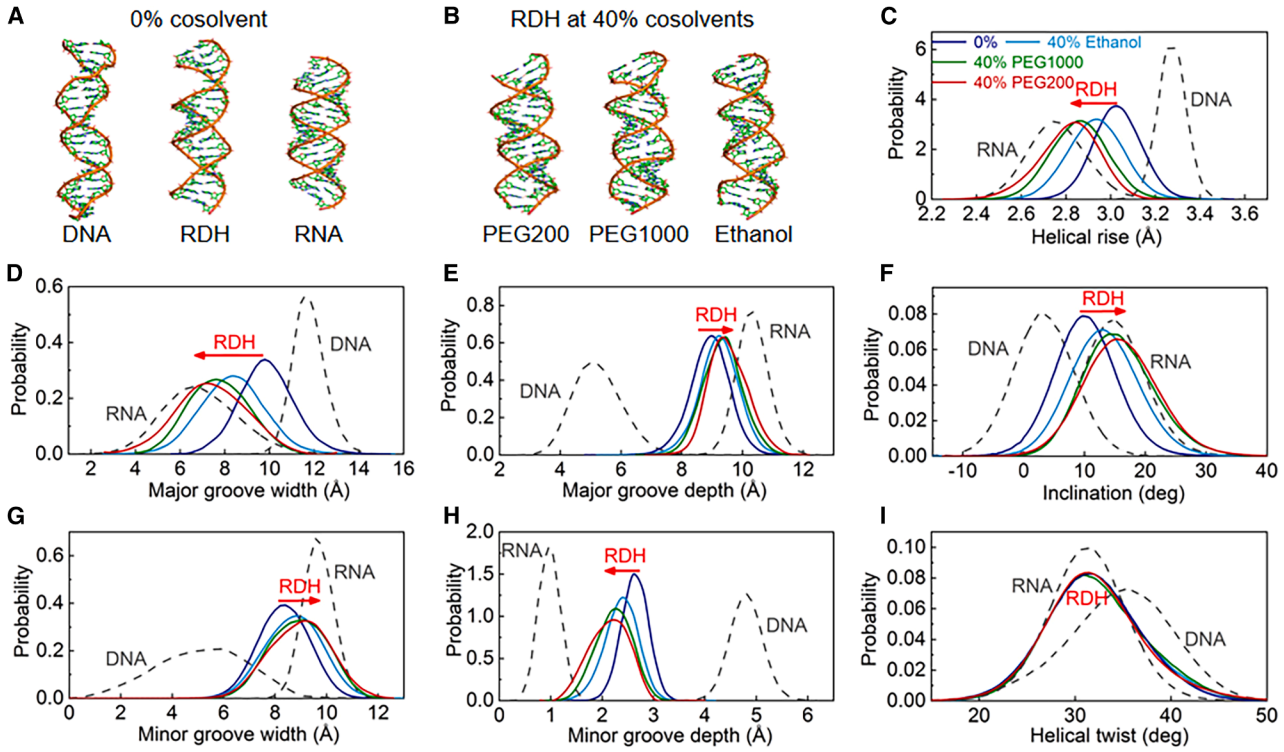


Figure 4. The structure and probability distributions of typical helical parameters for RDH, RNA, and DNA duplexes obtained from MD simulations. (A and B) The snapshots for RDH duplexes without and with added cosolvents at ∼500 ns and for the standard RNA and DNA duplexes as references. (C–I) The probability distributions of helical parameters including helical rise (C), the width and depth of the major groove (D and E), inclination (F), width and depth of the minor groove (G and H), and helical twist (I). Here, the distributions for RDH duplex are shown with solid lines (0% cosolvents, blue; 40% PEG200, red; 40% PEG1000, green; and 40% ethanol, cyan), and the distributions for RNA and DNA (dashed gray lines) duplexes without added cosolvents are shown as the references. The arrows represent the shift direction of the helical parameter with the addition of 40% cosolvents.

concentration is increased from 0% to 40%; see Figures 3C and 3D. Thus, our MD simulations quantitatively agree with the MT measurements on the effects of the cosolvents used on P and L_c of the RDH duplex.

Secondly, we calculated the twist-stretch coupling parameter dL/dN for the RDH duplex in the presence of PEG200, PEG1000, and ethanol over the concentration range of 0%–40% through the relationship between helical rise and helical twist (see materials and methods). We found that the used cosolvents all cause an apparent RNA-like positive twist-stretch coupling of the RDH duplex reflected by the decreases of dL/dN from ∼−0.10 to ∼−0.54 nm/turn for 40% PEG200, to ∼−0.50 nm/turn for 40% PEG1000, and to ∼−0.46 nm/turn for 40% ethanol, while 40% ethanol increases dL/dN for both RNA and DNA duplexes from our simulations as shown in Figures 3E and 3F. Our MD calculations of dL/dN quantitatively confirm the interesting findings from the MT measurements on the effect of cosolvents on the twist-stretch coupling of an RDH duplex, including PEG200, PEG1000, and ethanol.

Therefore, our extensive MD calculations nearly reproduced the MT measurements and verified the surprising findings that the cosolvents, including PEG200, PEG1000, and ethanol, strikingly cause a sharp bend softening for the RDH duplex and an apparently (RNA-like) positive twist-stretch coupling.

More flexible and more A-form-like structure of RDH duplex in cosolvent environments

To understand such interesting findings for the RDH duplex in the cosolvent environments, we analyzed the structure and helical parameters for the RDH duplex at typical concentrations of PEG200, PEG1000, and ethanol and used RNA and DNA duplexes without added cosolvents as references.[13,29,53] Firstly, based on the MD trajectories, we quantitatively characterize the extensive helical parameters of the RDH duplex, including the width and depth of the major groove, the width and depth of the minor groove, inclination, helical rise, and helical twist, which are key parameters to characterize the A-form and B-form helical structures.[7,13,16]

As shown in Figure 4, and Table S2, several aspects can be inferred with the addition of 40% cosolvent: (1) the fluctuations of helical parameters for the RDH duplex become more pronounced and (2) the values of the helical parameters shift from the near-intermediate values between A-form RNA and B-form DNA duplexes toward the values closer to those of the RNA duplex. These indicate the more

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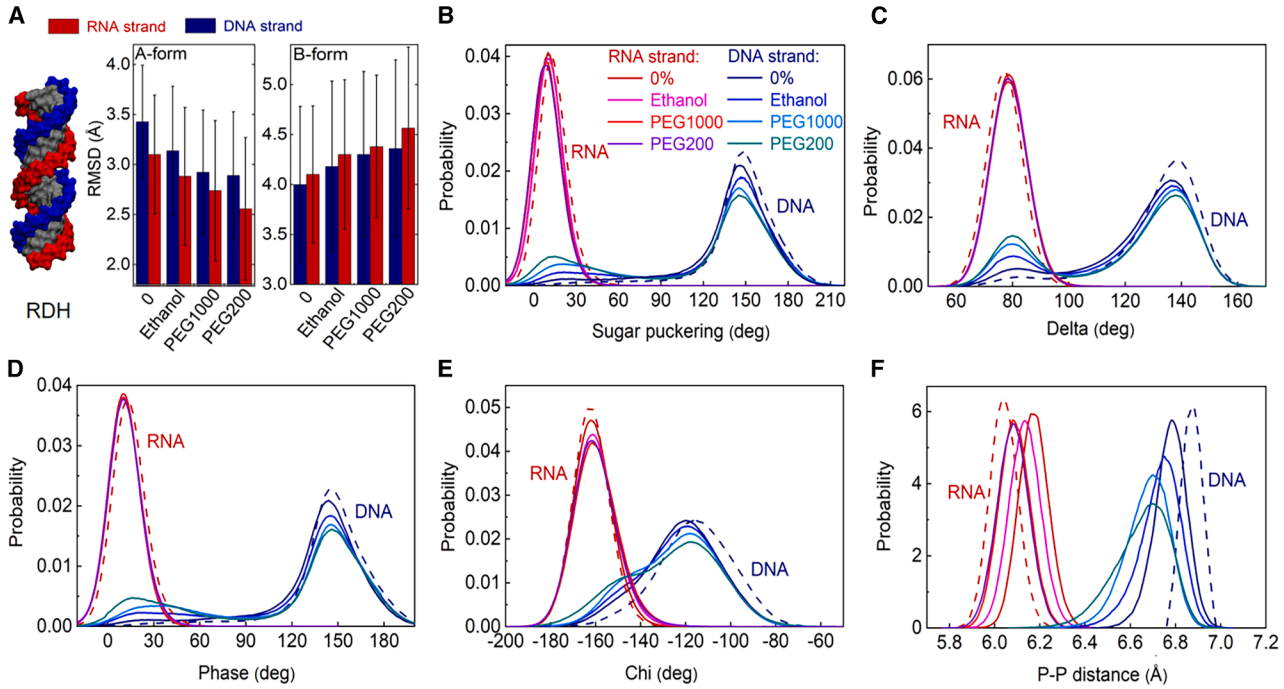


Figure 5. Structural characteristics of RNA and DNA strands in the RDH duplex from MD simulations. (A) Root-mean-square deviations (RMSDs) of RNA and DNA strands in the RDH duplex without cosolvents and with 40% cosolvents (PEG200, PEG1000, and ethanol), calculated with the canonical A-form (left) and B-form (right) strands as the reference structures. The error bars denote the standard deviations. (B–F) The probability distributions of structural parameters for the RNA and DNA strands in the RDH duplex without cosolvents and with 40% cosolvents including PEG200, PEG1000, and ethanol (solid lines), including sugar pucker angle (B), delta angle (C), phase angle (D), chi angle (E), and P-P distance (F). Here, the probability distributions of the corresponding RNA strand in the RNA duplex (dashed red line) and the corresponding DNA strand in the DNA duplex (dashed blue line) without added cosolvents are shown as references.

flexible and RNA-like structure of the RDH duplex with the addition of high cosolvents, as illustrated in Figures 4A and 4B. Specifically, the major groove width of the RDH duplex decreases from ∼10.0 to ∼7.3 Å for 40% PEG200, to ∼7.6 Å for 40% PEG1000, and to ∼8.6 Å for 40% ethanol, respectively, visibly closer to the value of ∼7.0 Å for the RNA duplex, as shown in Figure 4D. Moreover, the standard deviation of the major groove width of the RDH duplex increases from ∼1.2 to ∼1.6 Å for 40% PEG 200, to ∼1.5 Å for 40% PEG1000, and to ∼1.5 Å for 40% ethanol, which are also closer to that (∼1.7 Å) of the RNA duplex, respectively. Here, we explicitly showed changes in major groove width and its fluctuation upon the addition of the cosolvents since these two quantities are signature parameters for characterizing A-form/B-form structures and mechanical properties.[7,13,16,75] Therefore, the above analyses on helical parameters suggested that the helical structure of the RDH duplex becomes more A-form-like and more flexible in the presence of the cosolvents.

In order to reveal why the structure of the RDH duplex becomes closer to A-form RNA in cosolvent environments, we would analyze the structures of the two strands separately rather than the whole duplex because DNA and RNA strands can adopt different structures, as suggested in previous MD simulations.[7] Firstly, we calculate the root-mean-square deviations (RMSDs) for the two strands of the RDH duplex with the canonical A-form and B-form strands as reference structures. As shown in Figure 5A, 40% PEG200, PEG1000, and ethanol all decrease the mean RMSDs of the two strands relative to the canonical A-form structure while increasing the RMSDs relative to the canonical B-form one, indicating that the DNA strand and RNA strand in the RDH duplex both shift to an A-form structure and deviate from a B-form one in the cosolvent environments. Secondly, we calculated the rigid body parameters and the torsion angles, including sugar pucker angle, delta angle, phase angle, chi angle, and P-P distance, respectively, which are key parameters to characterize the structures of single DNA and RNA strands in an RDH duplex.[7,8,76] As shown in Figures 5B–5F, the parameters of DNA and RNA strands in the RDH duplex are almost identical to the corresponding DNA strand in the pure DNA duplex and the corresponding RNA strand in the pure RNA duplex without the added cosolvents, respectively. However, with the addition of 40% PEG200, PEG1000 and ethanol, the parameters of the RNA and DNA strands in the RDH duplex exhibit the following major features: (1) the distributions for the RNA strand remained almost unchanged and generally appear narrower than those of the DNA strand, suggesting the RNA strand is more rigid than the DNA strand in the RDH duplex, and (2) the parameter distributions of DNA strand become closer to those of RNA duplex reflected by the distribution shifts toward those of RNA strand and the additional

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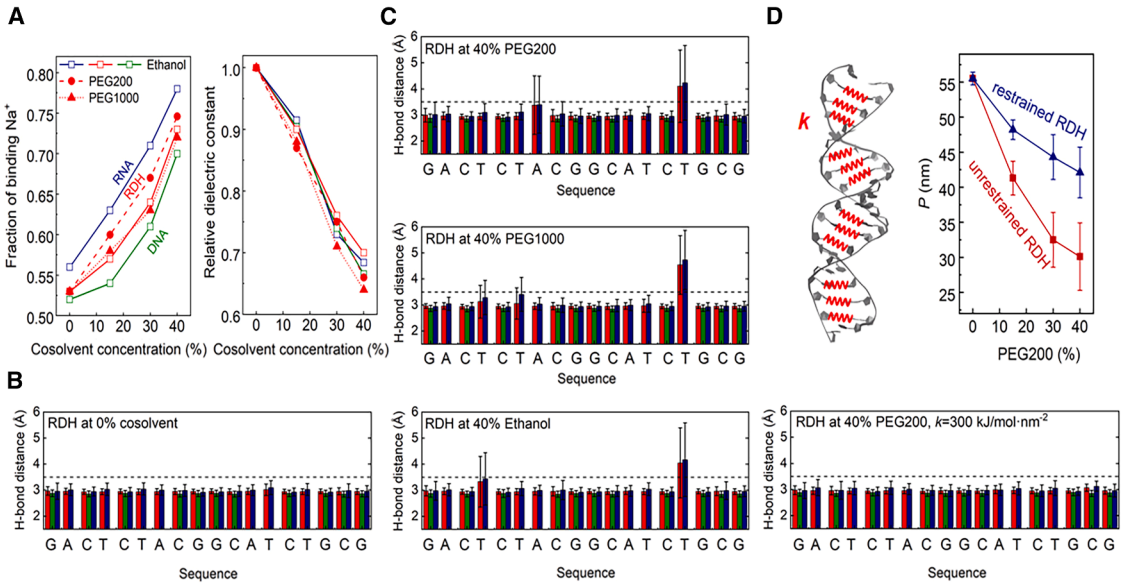


Figure 6. Effect of cosolvent on the ion binding, solvent dielectric constant, and distribution of hydrogen bond length for RDH duplex. (A) The fractions of ion binding as functions of cosolvent concentration. The relative dielectric constant ($\varepsilon_{eff}/\varepsilon_w$) of the solutions with cosolvents to that of pure water as functions of cosolvent concentration. The open blue, red, and green squares denote RNA, RDH, and DNA duplexes in the presence of ethanol, respectively, whereas the filled red circles and triangles denote RDH duplexes in the presence of PEG200 and PEG1000, respectively. (B and C) The distance distribution of hydrogen bonds along RDH duplex without cosolvents (B) and with 40% cosolvents including PEG200, PEG1000, and ethanol (C). The red, green, and blue bars denote individual hydrogen bond lengths within the respective base pairs, and the error bars denote the standard deviations. (D) Schematic illustration for the MD simulations with artificially added spring constraints for hydrogen bonds (left), P as a function of PEG200 concentration with and without the added constraints (right), and the distribution of hydrogen bond length with spring constraints along RDH duplex at 40% PEG200 (bottom). Red and blue curves denote the cases without and with the constraints, respectively. Here, the spring constant k for the constraints used was set to 300 kJ/mol · nm^2. The error bars in panel D denote the standard deviations.

distribution appearances of some parameters at the distributions of RNA strand. These indicate that the cosolvents cause a pronounced bimodal behavior of the DNA strand for some parameters and promote a tendency toward B-to-A transition for the DNA strand in the RDH duplex. Please see the distributions of other torsional angles in Figure S7. Furthermore, we analyzed the distributions of cosolvents around the RDH duplex as shown in Figure S8. With increasing cosolvent concentration, the number density of water molecules around the RDH slightly decreased, whereas PEG and ethanol were slightly depleted near the duplex. This supports a cosolvent-dehydration-driven conformational shift rather than the formation of a dense PEG layer or strong specific PEG-RDH interactions.

Therefore, our MD-based analyses reveal that the RDH duplex adopts a more A-like and flexible conformation in the cosolvent environments, mainly due to the tendency of the B-to-A transition in the DNA strand of the RDH duplex promoted by the cosolvents.

Sharp bend softening of RDH duplex is mainly ascribed to hydrogen-bond disruptions

To reveal why the cosolvents induce a sharp bend softening of the RDH duplex, we first analyzed the ion distributions around the RDH duplex in different cosolvents, including PEG200, PEG1000, and ethanol, at a typical 40% concentration, given the polyanionic nature of nucleic acids.[7,27,30,31,77–81] As shown in Figure 6A, the addition of the 40% cosolvents leads to an apparent increase in ion binding by ∼39% for PEG200, by ∼36% for PEG1000, and by ∼38% for ethanol, which is similar to the enhanced ion binding for RNA (∼39%) and DNA (∼35%) duplexes by 40% ethanol. Such apparently enhanced ion binding is caused by the decrease in effective dielectric constant of bulk solvent with the addition of the cosolvents with lower polarity, as shown in Figure 6A,[16,68] which would cause a more flexible RDH duplex through greatly weakening the electrostatic repulsion of the RDH duplex and enhancing the strand fluctuations in an analogous way to RNA and DNA duplexes in 40% ethanol solution.[16] However, as shown in Figure 3A, the addition of 40% cosolvent causes a sharp softening in bending for the RDH duplex, notably more than the softening effect on RNA and DNA duplexes by 40% ethanol. Besides, it is not strange that P of the RDH is smaller at high PEG200 than that at high PEG1000. This is attributed to the more enhanced ion binding promoted by high PEG200 than high PEG1000 and the associated stronger structure fluctuation (see Figures 6A and 4). Such higher ion binding may be caused by the stronger hindering effect of the larger size of PEG1000 than PEG200.

To further understand the sharp bend softening of the RDH duplex by the cosolvents, we examined the hydrogen

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bonds along the RDH duplex with the addition of 40% cosolvents, including PEG200, PEG1000, and ethanol, since the above-shown structure mismatch between the RNA and DNA strands and enhanced strand fluctuation by the cosolvents may affect the base-pairing hydrogen bonds of the RDH duplex. As shown in Figures 6B, 6C, and S9, there are certain disrupted hydrogen bonds generally occurring at A-T base pairs rather than C-G base pairs along the RDH duplex at 40% PEG200, PEG1000, and ethanol, whereas the hydrogen bonds along RNA and DNA duplexes are negligibly affected by 40% ethanol. The local hydrogen-bond disruption at certain AT base pairs by the 40% cosolvents would cause the sharp bend softening of the RDH duplex in the cosolvent environments. Such apparently unstable base pairing in the RDH duplex compared to RNA and DNA duplexes should be caused by a structure mismatch between the B-form DNA strand and the A-form strand in the RDH duplex,[7] which is also reflected by the nearest-neighbor thermodynamic parameters.[32] The local hydrogen bonds break primarily at the base pairs where thermodynamic stability is most markedly reduced with the addition of cosolvents, according to nearest-neighbor thermodynamic parameters for RDH duplexes.[32] Besides, local hydrogen bonds along the RDH duplex at 15% and 30% cosolvent concentrations are also slightly weakened, as shown in Figure S9. To confirm the role of hydrogen-bond disruptions in the sharp softening of the RDH, we performed additional MD simulations for the RDH with artificially added harmonic constraints for hydrogen bonds (see Figure 6D). The additional MD simulations showed that the recovery of disrupted hydrogen bonds substantially increased the bending persistence length P, especially at high PEG200, e.g., at 40% PEG200, the P of the RDH increased from ∼30.1 to ∼42.7 nm, as shown in Figure 6D. This confirms the essential role of hydrogen-bond disruption in causing the pronounced bending softening of the RDH.

To explore the major contribution of high cosolvents to such apparent softening, we set up a model system for additional MD simulations in which the atomic charges of water molecules were reduced proportionally to effectively decrease the solution polarity, and no cosolvents were added (see Figure S10). Such MD systems mimic the reduced dielectric environment by excluding the exclusion volume effect from high cosolvents.[53] The MD simulations showed that when the effective dielectric constant decreased to that of 40% cosolvents, the P of the RDH duplex decreased sharply from ∼55.5 to ∼36.5 nm, a value very close to that at 40% cosolvents, accompanied by enhanced strand fluctuation due to increased ion binding and local hydrogen-bond disruptions, in analogy with those in the presence of high cosolvents (see Figure S10). This indicates that the decreased effective dielectric constant of solutions by high cosolvents makes a major contribution to the sharp softening of the RDH in bending, since a similar decrease in the effective dielectric constant leads to a similar apparent decrease in P for both the model system and the system with high cosolvents, including 40% PEG1000 and PEG200.

Moreover, to directly figure out the effect of volume exclusion by the cosolvents, we set up another model system for MD simulations in which the atom charges of water and PEG200/PEG1000 molecules were proportionally enlarged to increase the polarity of the solutions to that of solutions without added cosolvents. Thus, in such MD systems, the volume-exclusion effect of cosolvents was kept while the effect of cosolvents on solution polarity was diminished. The MD simulations showed that when the effective dielectric constant was restored to nearly that without cosolvents, the P of the RDH duplex was ∼54.7 nm at 40% PEG200 and ∼58.6 nm at 40% PEG1000, close to the value without cosolvents (∼55.5 nm). This was accompanied by reduced ion binding and largely restored hydrogen bonds (see Figure S10). These results further indicate that the pronounced bending softening of the RDH is primarily driven by the cosolvent-induced reduction in the solvent polarity, whereas the contribution from excluded-volume interactions alone is relatively weak. The RDH duplex focused in this work is apparently different from the DNA duplex addressed in recent experiments focused on the DNA duplex in the presence of 20% diglyme[82] since there is a conformation mismatch between the A-form RNA and B-form DNA strands in the RDH. Such a conformation mismatch, together with the above-shown increased ion binding and the enhanced strand fluctuation by high cosolvents, leads to local hydrogen-bond disruptions and consequently sharp softening of the RDH in bending. Such a mechanism can be understood since the electrostatic repulsion is still rather strong within the RDH with only partial charge neutralization (binding-ion fraction ∼53%), and the relatively small size of the cosolvents used may also limit the volume-exclusion contribution of the cosolvents.

Therefore, the sharp bend softening of the RDH duplex in the cosolvent environments is mainly attributed to local hydrogen-bond disruption jointly caused by the structure mismatch between the RNA and DNA strands and enhanced strand fluctuation due to increased ion neutralization through decreasing solution polarity.

A-form-like deformation pathway causes RNA-like twist-stretch coupling of RDH duplex in cosolvent environments

To understand the RNA-like positive twist-stretch coupling ($dL/dN < 0$) of the RDH duplex caused by cosolvents from a negligible one ($dL/dN \sim 0$), we first examined the relationships between the helical rise and twist through the major groove width and helical radius for the RDH duplex without cosolvents and with 40% cosolvents, including PEG200, PEG1000, and ethanol, in analogy to the previous analyses.[16,28,29] Furthermore, we examined the relationships

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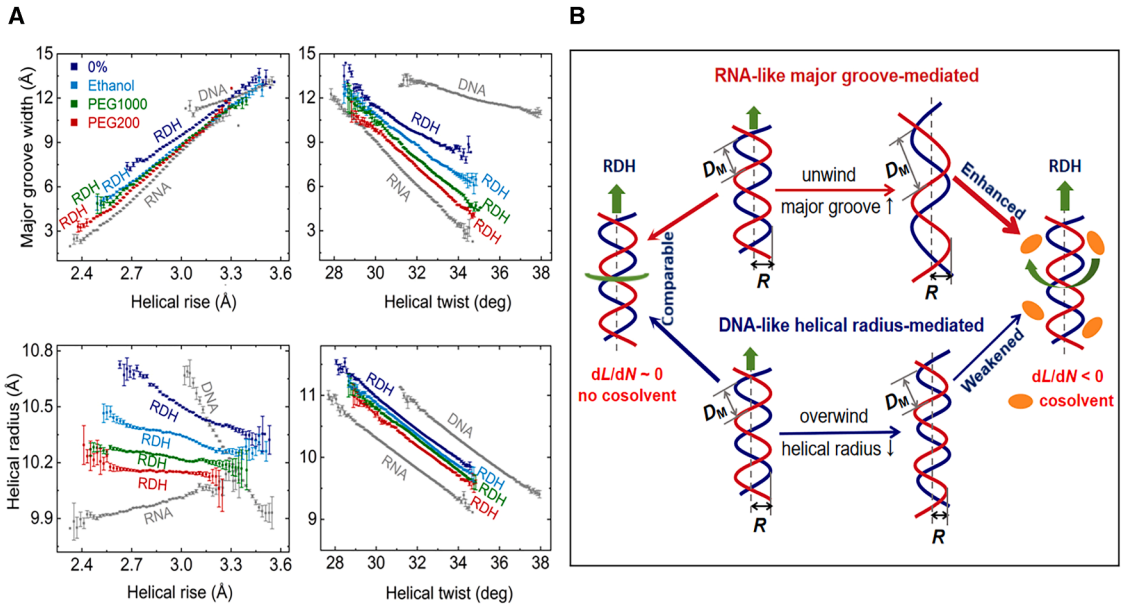


Figure 7. Relationships between typical helical parameters for RDH duplex without and with 40% cosolvents, including PEG200, PEG1000, and ethanol, from MD simulations. (A) Major groove width versus helical rise/helical twist (top) and helical radius versus helical rise/helical twist (bottom). (B) A schematic illustration of two competitive deformation pathways for twist-stretch coupling of RDH duplex, including the RNA-like major groove-mediated pathway and the DNA-like helical radius-mediated pathway. Here, R and D_M denote the helical radius and major groove width, respectively. The corresponding RNA (gray line) and DNA (gray line) duplexes without added cosolvents are shown as the references. The error bars denote the standard errors.

between the helical rise and twist using key base-pair parameters at the base-pair level and the signature parameter of sugar puckering at the strand level. Here, we used RNA and DNA duplexes without added cosolvents as the references for the analyses.

Major groove width-mediated pathway overwhelms helical radius-mediated one at helix level

As shown in Figure 7, in the absence of the cosolvents, the major groove width modestly and positively correlates with helical rise while modestly and negatively correlating with helical twist, causing a major groove-mediated negative twist-stretch coupling parameter dL/dN for the RDH duplex. Simultaneously, there is a strong/negative correlation between helical radius and helical twist and a modest/negative correlation with helical rise, causing a helical radius-mediated positive dL/dN for the RDH duplex. The opposite effect of the major groove-mediated pathway and helical radius-mediated pathway leads to a negligible twist-stretch coupling for the RDH duplex without cosolvents. However, as shown in Figure 7, with the addition of 40% cosolvents such as PEGs and ethanol, the correlations between major groove width and helical rise/helical twist are both enhanced, causing more strongly positive twist-stretch coupling (more negative dL/dN) mediated by major groove width for the RDH duplex. Meanwhile, the correlations between helical radius and helical rise of the RDH duplex become rather weak at 40% PEG200, PEG1000, and ethanol, inducing significantly weakened negative twist-stretch coupling (weakly positive dL/dN) mediated by helical radius for the RDH duplex. Thus, the enhanced (strong) major-groove mediated pathway for negative dL/dN and significantly weakened (weak) helical radius-mediated one for positive dL/dN together lead to an apparently positive (RNA-like) twist-stretch coupling (apparently negative dL/dN) for the RDH duplex in the presence of 40% PEG200, PEG1000, and ethanol, which is similar to the deformation pathway for the twist-stretch coupling of RNA duplex, as shown in Figure 7.[28,29,53,83]

Roll-mediated pathway overwhelms x-displacement/slide-mediated one at base-pair level

To further elucidate the RNA-like twist-stretch coupling of the RDH duplex in the cosolvent environments at the base-pair level, we analyzed the Pearson correlation coefficients (PCCs) between the key helical parameters of major groove width/helical radius and all important microscopic base-pair structure parameters[28,84] (see Tables S3 and S4). For the major groove, which mediates the RNA-like twist-stretch coupling pathway, the most apparent correlation occurs between major groove width and roll for RDH and RNA duplexes (with PCCs > $\sim$−0.75), whereas for the helical radius, which mediates the DNA-like twist-stretch coupling pathway, the most visible correlations occur between helical radius and x-displacement/slide (PCC >-$\sim$−0.60). In the following, we examined the deformation pathways for twist-stretch coupling of the RDH duplex through roll, x-displacement, and slide at the base-pair level

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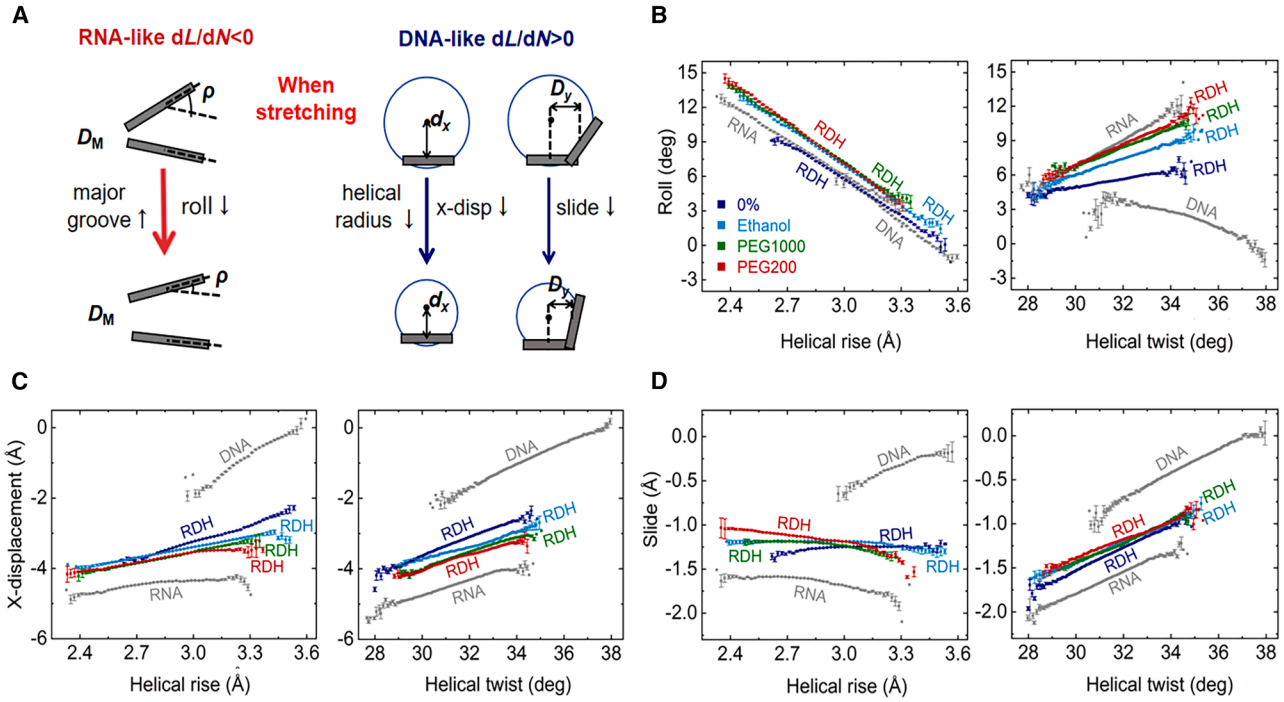


Figure 8. Relationships of major groove width and helical radius with typical base-pair parameters for RDH duplex without and with 40% cosolvents, including PEG200, PEG1000, and ethanol, from MD simulations. (A) An illustration of the relationship between major groove width and roll (left) and an illustration of the relationships between helical radius and x-displacement/slide (right). Here, D_M, ρ, d_x, and D_y denote major groove width, roll, x-displacement, and slide, respectively. (B–D) Roll/x-displacement/slide versus helical rise/helical twist. The corresponding RNA (gray line) and DNA (gray line) duplexes without added cosolvents are shown here as references. The error bars denote the standard errors.

beyond the major groove and helical radius at the helix level (see Figure 8).

For RDH duplexes without added cosolvents, roll is strongly and negatively correlated with helical rise while positively correlated with helical twist, indicating an RNA-like roll-mediated pathway for dL/dN < 0 (positive twist-stretch coupling), whereas x-displacement/slide are strongly and positively correlated with helical twist and moderately/positively correlated with helical rise, causing a DNA-like x-displacement/slide-mediated pathway for dL/dN > 0 (negative twist-stretch coupling). The two comparable and opposite deformation pathways lead to a negligible twist-stretch coupling for the RDH duplex in the absence of cosolvents. With the addition of 40% cosolvents including PEG200, PEG1000, and ethanol, the roll-mediated pathway is visibly enhanced, while the x-displacement/slide-mediated one is visibly weakened, which leads to a globally RNA-like pathway and an RNA-like positive twist-stretch coupling (dL/dN < 0) for the RDH duplex in the presence of 40% cosolvents.

C3′-endo-mediated pathway overwhelms C2′-endo-mediated pathway at strand level

To further examine the origin of the RNA-like positive twist-stretch coupling for the RDH duplex at the strand level, we analyzed the RNA and DNA strands separately rather than the whole duplex. The opposite deformation behaviors of RNA and DNA strands upon stretching can be traced to their fundamental difference in sugar puckering, which could distinguish the essential difference between A-form and B-form duplexes.[48,84] As shown in Figure 9, the RNA strand favors C3′-endo sugar puckering with an angle around ∼11°, and the DNA strand favors C2′-endo sugar puckering with an angle around ∼140° in the absence of cosolvents. In the following, we examined the correlations of sugar pucker angle with the key base-pair parameters of roll, x-displacement, and slide, which play major roles in mediating deformation pathways for the twist-stretch coupling of the RDH duplex.

As shown in Figure 9, the C3′-endo sugar puckering for the RNA strand mainly dominates the positive twist-stretch coupling (dL/dN < 0) through the strong and positive coupling between roll and sugar pucker angle, whereas the C2′-endo sugar puckering for the DNA strand mainly dominates the negative twist-stretch coupling (dL/dN > 0) through the apparent correlations between x-displacement/slide and sugar pucker angle. The opposite effects of two sugar puckering-mediated pathways lead to a negligible twist-stretch coupling for the RDH duplex without added cosolvents. With the addition of 40% cosolvents including PEG200, PEG1000, and ethanol, the C3′-endo-mediated pathway (dL/dN < 0) is slightly enhanced by

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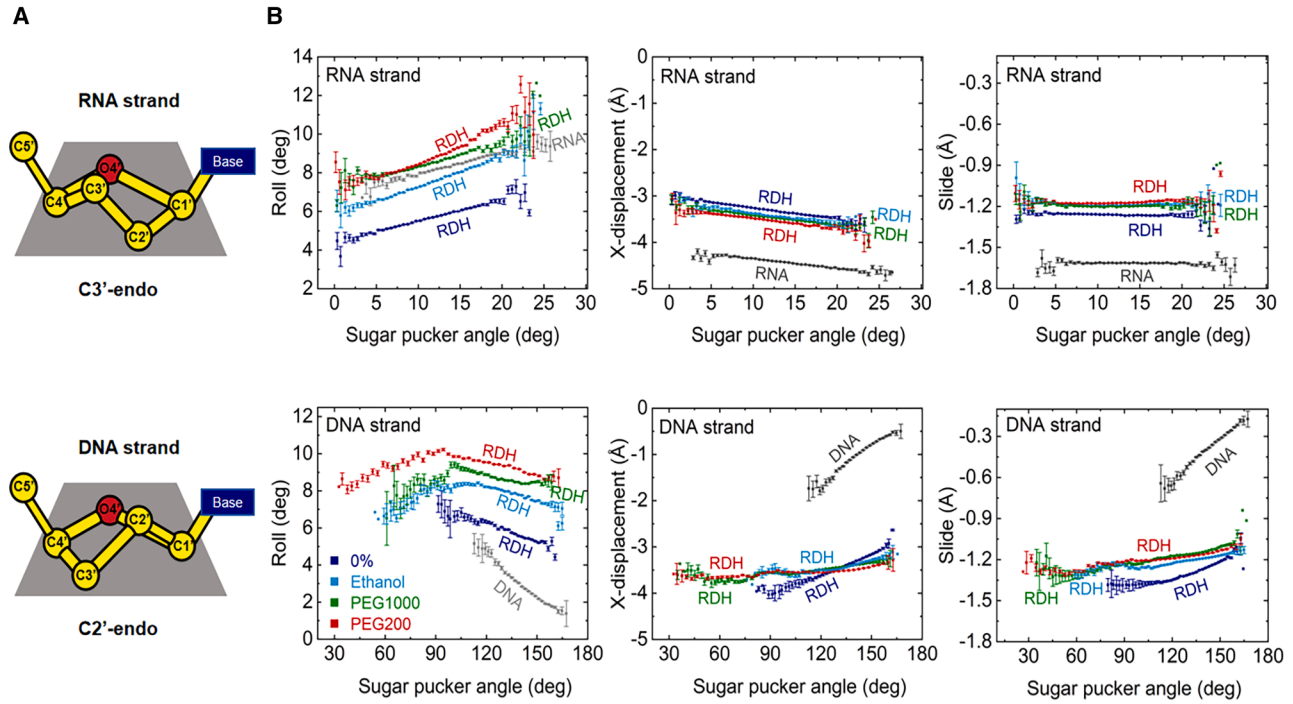


Figure 9. Relationships between sugar pucker angle and typical base-pair parameters for RDH duplex without and with 40% cosolvents, including PEG200, PEG1000, and ethanol, from MD simulations. (A) A schematic illustration of the two signature sugar puckering modes of RNA (C3′-endo) and DNA (C2′-endo) strands. (B) Roll/x-displacement/slide versus sugar pucker angle for RNA (top) and DNA (bottom) strands for RDH duplex. Here, the corresponding RNA strand in the RNA duplex (gray line) and the corresponding DNA strand in the DNA duplex (gray line) without added cosolvents are shown as references. The error bars in (B) denote the standard errors.

the increased coupling between roll and sugar pucker angle, whereas the C2′-endo-mediated pathway ($dL/dN > 0$) is weakened by the decreased coupling between x-displacement/slide and sugar pucker angle, leading to a global RNA-like positive twist-stretch coupling ($dL/dN < 0$) for the RDH duplex.

Therefore, the RNA-like positive twist-stretch coupling of the RDH duplex in cosolvent environments is attributed to a more A-form RNA-like deformation pathway mainly through the enhanced RNA-like major groove width (roll)-mediated pathway and the weakened DNA-like helical radius (x-displacement/slide)-mediated pathway induced by the cosolvents. At the strand level, the RNA-like pathway is enhanced by the increased coupling between roll and sugar pucker angle, whereas the DNA-like pathway is weakened by the decreased coupling between x-displacement/slide and sugar pucker angle, leading to the global RNA-like twist-stretch coupling for the RDH duplex by the cosolvents.

CONCLUSION

In conclusion, we explored the structure and mechanics of the RDH duplex in the cosolvent environments by combining high-precision MT experiments with all-atom MD simulations, using PEG200, PEG1000, and ethanol as representative model cosolvents over a wide concentration range. First, we found that the cosolvents sharply soften the RDH duplex in bending and cause an apparently RNA-like positive twist-stretch coupling of the RDH duplex. Second, the RDH duplex adopts a more A-form-like and more flexible structure in cosolvent environments, which is attributed to the more rigid RNA strand than the DNA strand and the shift of the DNA strand from B-form to A-form caused by the cosolvents. Specifically, the 40% cosolvent causes a sharp decrease in bending persistence length P from ∼54 to ∼27–35 nm (∼35%–50% decrease), which is mainly ascribed to the local hydrogen-bond disruption caused by the structure mismatching between the RNA and DNA strands in the RDH duplex and enhanced strand fluctuation associated with increased ion binding. Meanwhile, the twist-stretch coupling parameter dL/dN for the RDH duplex decreases from ∼0 to −0.56−-0.45 nm/turn when 40% cosolvents are added, indicating a shift toward an RNA-like twist-stretch coupling. Such a shift is attributed to the enhancement of the major groove-mediated pathway and the weakening of the helical radius-mediated pathway. At the base-pair level, two deformation pathways are mainly mediated by roll and x-displacement/slide, respectively, and are ultimately ascribed to the distinct sugar-puckering preferences: the C3′-endo-mediated pathway for RNA strands and the C2′-endo-mediated pathway for DNA strands. These

findings may be relevant to intracellular environments whose effective dielectric constants are estimated to be ∼20–60, values lower than that of bulk water.[68] The comparable dielectric response of the high cosolvent systems suggests that the sharp cosolvent-mediated softening and RNA-like twist-stretch coupling of the RDH duplex may provide some insights into nucleic acid mechanics in intracellular environments, although these simplified systems do not capture specific protein-nucleic acid or quinary interactions. Therefore, our findings provide a comprehensive understanding of the RDH duplex in cosolvent environments and may be helpful for elucidating its biological functions in cells and for the rational design of RDH-based nanostructures.

DATA AND CODE AVAILABILITY

The data underlying this article are available in the article and its online supplemental information.

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AUTHOR CONTRIBUTIONS

Z.-J.T., X.-H.Z., C.-C.Z., and Y.-L.C. designed the research; Y.-L.C. and X.-H.Z. performed the experimental measurements; C.-C.Z. and Z.-J.T. performed the simulations; Z.-J.T., C.-C.Z., Y.-L.C., Z.W., and L.D. analyzed the data; and C.-C.Z., Y.-L.C., L.D., X.-H.Z., and Z.-J.T. wrote the manuscript. All authors discussed the results and reviewed the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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